Peilin (Leo) Chen

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Research Interests

- **AI Chip Based on Computing-in-Memory**
- **Algorithm-Hardware Co-Design**
- **Neuromorphic Computing**
- **Computer Architecture**

Education

University of Virginia — **Aug. 2024 – Present**
Ph.D. in Electrical Engineering — *Advisor: Dr. Xiaoxuan Yang*

University of Virginia — **Aug. 2024 – Dec. 2026**
M.E. in Electrical Engineering — *Advisor: Dr. Xiaoxuan Yang*

Xidian University — **Sept. 2020 – Jun. 2024**
B.E. in Integrated Circuit Design and Integration System — *Advisor: Dr. Kang Li*

Experience

University of Virginia — **Aug. 2024 – Present**
Research Assistant — *Advisor: Dr. Xiaoxuan Yang*

Nanyang Technological University — **Jul. 2023 – Aug. 2023**
Visiting Student — *Mentor: Dr. Goh Wang Ling, Dr. Tony Tae-Hyoung Kim*

Hong Kong University of Science and Technology (Guangzhou) — **Jun. 2023 – Jul. 2023**
Research Assistant — *Mentor: Dr. Hongwu Jiang*

Shenzhen Xiyun Digital Technology Co., Ltd — **Nov. 2021 – Jan. 2022**
Technology Intern — *Mentor: Mr. Ziye Lv*

Publications

Conference Proceedings

- **Peilin Chen** and Xiaoxuan Yang, "Lonic: Algorithm-Hardware Co-Design for Energy-Efficient Fully Local Online SNN Training with INT4 Precision," *In ACM/IEEE International Conference on Computer-Aided Design (ICCAD), San Jose, California, USA, 2026.* **(Rank First in the Track)**
- **Peilin Chen** and Xiaoxuan Yang, "SpikON: A Dual-Parallel and Efficient Accelerator for Online Spiking Neural Networks Learning," *In ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED), Evanston, Illinois, USA, 2026.* **(Invited for TVLSI Extension)**
- **Peilin Chen** and Xiaoxuan Yang, "SpiKint: Native Full-Integer Spiking Neural Networks Training with an Efficient CIM-based Accelerator," *In ACM Great Lakes Symposium on VLSI (GLSVLSI), Canandaigua, NY, USA, 2026.*
- Xiaoxuan Yang, **Peilin Chen**, Tergel Molom-Ochir, and Yiran Chen, "End-to-End Transformer Acceleration Through Processing-in-Memory Architectures," *In IEEE International Conference on Microelectronics (ICM), Cairo, EGYPT, 2025.*
- **Peilin Chen** and Xiaoxuan Yang, "Titanus: Enabling KV Cache Pruning and Quantization On-the-Fly for LLM Acceleration," *In ACM Great Lakes Symposium on VLSI (GLSVLSI), New Orleans, USA, 2025.* **(Rank First in the Track) (Best Paper Award)**

- **Peilin Chen** and Xiaoxuan Yang, "Optimizing and Exploring System Performance in Compact Processing-in-Memory-Based Chips," *In IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS), Bordeaux, FRANCE, 2025.*

Journal Articles

- **Peilin Chen**, Kevin Skadron, and Xiaoxuan Yang, "SpiKint-v2: Temporal-Gradient-Free Native Integer SNN Training with an Efficient CIM Accelerator," *In IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026.* (Under Review)
- Hanyuan Gao, Tergel Molom-Ochir, **Peilin Chen**, Xiaoxuan Yang, and Hai "Helen" Li, "Toward Memory-Efficient Transformer Inference for Processing-in-Memory Systems," *In IEEE Transactions on Circuits and Systems for Artificial Intelligence (TCASAI), 2026.* (Under Review)
- Jiacheng Cao, Wei Xiong, Jie Lu, **Peilin Chen**, Jian Wang, Jinmei Lai, and Miaoqing Huang, "An optimized EEGNet processor for low-power and real-time EEG classification in wearable brain-computer interfaces," *In Microelectronics Journal, 2024.*

Selected Awards and Honors

- **Best Paper Award,** ACM Great Lakes Symposium on VLSI (GLSVLSI), 2025
- **DAC Young Fellow,** ACM/IEEE Design Automation Conference (DAC), 2025
- **Provost's Fellowship,** University of Virginia, 2024
- **Excellent Graduation Thesis,** Xidian University, 2024
- **National Third Prize,** the 25th China Robot and Artificial Intelligence Competition, 2023
- **National Second Prize,** the 6th China College Integrated Circuit Competition, 2022
- **Meritorious Winner,** COMAP Mathematical Contest in Modeling, 2022
- **Excellent,** National Undergraduate Innovation and Entrepreneurship Training Program, 2022
- **National Third Prize,** Embedded System Design Invitational Contest (Intel Cup), 2022
- **National Second Prize,** China Undergraduate Mathematical Contest in Modeling, 2021
- **National Third Prize,** National Undergraduate FPGA Innovation Design Competition, 2021
- **Provincial Second Prize,** Mathematics Competition of Chinese College Student, 2021
- **Provincial Second Prize,** Chinese Collegiate Computing Competition, 2021

Presentations

Oral Presentation

- **ISLPED**, Evanston, Illinois. Aug. 2026
  SpikON: A Dual-Parallel and Efficient Accelerator for Online Spiking Neural Networks Learning
- **GLSVLSI**, Canandaigua, New York. Jun. 2026
  SpiKint: Native Full-Integer Spiking Neural Networks Training with an Efficient CIM-based Accelerator
- **GLSVLSI**, New Orleans, Louisiana. Jun. 2025
  Titanus: Enabling KV Cache Pruning and Quantization On-the-Fly for LLM Acceleration

Poster Presentation

- **DAC**, San Francisco, California. Jun. 2025
  From PIM Architecture to LLM Acceleration: Co-Designing Efficient AI Systems

Services and Mentoring

Journal Reviewer

- IEEE Transactions on Computers (TC), 2026
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2026

Research Mentorship

- Daniel Cao, Undergraduate student at UVA, Electrical and Computer Engineering
- Junting Huo, Undergraduate student at UVA, Electrical and Computer Engineering

Skills and Others

Programming Languages: Verilog, SystemVerilog, Python, C++, C, TCL, Matlab, LaTex, Markdown
EDA Tools: Cadence Innovus, Cadence Virtuoso, Synopsys Design Compiler, Synopsys IC Compiler, Synopsys Library Compiler, Synopsys VCS, Synopsys Verdi, Siemens Modelsim, Xilinx Vivado
ML Frameworks: PyTorch, TensorFlow